

A rapid reliability estimation method for directed acyclic lifeline networks with statistically dependent components

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ARTICLE INFO

Article history:

Received 21 January 2013

Received in revised form

13 November 2013

Accepted 28 November 2013

Available online 7 December 2013

Keywords:

Earthquake hazards

Lifeline network

Non-simulation-based method

Statistical correlation

System reliability

ABSTRACT

Lifeline networks, such as transportation, water supply, sewers, telecommunications, and electrical and gas networks, are essential elements for the economic and societal functions of urban areas, but their components are highly susceptible to natural or man-made hazards. In this context, it is essential to provide effective pre-disaster hazard mitigation strategies and prompt post-disaster risk management efforts based on rapid system reliability assessment. This paper proposes a rapid reliability estimation method for node-pair connectivity analysis of lifeline networks especially when the network components are statistically correlated. Recursive procedures are proposed to compound all network nodes until they become a single super node representing the connectivity between the origin and destination nodes. The proposed method is applied to numerical network examples and benchmark interconnected power and water networks in Memphis, Shelby County. The connectivity analysis results show the proposed method's reasonable accuracy and remarkable efficiency as compared to the Monte Carlo simulations.

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1. Introduction

Lifeline networks, such as transportation, water supply, sewers, telecommunications, and electrical and gas networks, are fundamental to our environment, socio-economic development, and quality of life. Although composed of redundant components, they are still susceptible to natural hazards as evidenced by historical earthquake events such as the Loma Prieta earthquake in the USA in 1989, the Northridge earthquake in California in the USA in 1994, the Hyogoken Nanbu earthquake in Japan in 1995, and the Chile earthquake in 2010. To prevent the spread of disaster and to minimize the effects of natural and man-made hazards to these lifeline network systems, it is essential to provide prompt and effective pre-disaster hazard mitigation strategies and post-disaster recovery efforts based on rapid system reliability assessment.

System reliability assessment of lifeline networks is often carried out by a node-pair connectivity analysis which determines the reliable state of a network as the connectivity between the origin and destination nodes without considering the network flow quantities. This node-pair connectivity analysis has been used widely in the post-hazard reliability analysis of lifeline networks because it adequately and efficiently describes the impact of disasters on lifeline disruptions. The connectivity is a necessary condition of more sophisticated flow-based analyses, and it is important to study connectivity particularly for more complex lifeline systems. In spite of the simplicity of

the node-pair connectivity analysis, this connectivity analysis is still challenging because lifeline networks are highly complex systems consisting of spatially distributed network components, intricate topology, uncertainties in actual conditions of network components and deterioration models, and interdependencies at the component and system levels [1]. Considering these complex natures, the reliability estimation of a node-pair connectivity problem is a difficult and complex task.

To overcome these challenges, many researchers have relied on Monte Carlo simulation (MCS) based methods because they are easy to implement. They just require the generation of random samples of hazard intensity measures and the corresponding uncertain status of network components with no need to identify complex connection system events [2]. However, the statistical nature of MCS based methods may need a large number of samples to achieve an acceptable level of convergence for very small or large probability estimation [3]. In addition, their computational efficiency depends on the number of nodes and links in a network as they use a path searching algorithm to check the connectivity between the origin and destination nodes for each sample [4]. Also, the random number generation process is computationally demanding especially when the components are statistically correlated as the process includes a matrix decomposition process such as the Cholesky decomposition. Depending on the nature of a given network, these challenges may hamper the rapid reliability estimation and prompt decision supports for post-disaster recovery efforts.

On the other hand, non-simulation based methods have been developed since the 1970s to calculate a node-pair connection probability by identifying the system events or sets without relying

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on random samples. They were developed based on Boolean algebra, set theory, Binary decision diagrams, network decomposition, recursive disjoint set identification, and minimal path identification [5–14]. They efficiently calculated the node-pair connection probability of general networks, and provided a high computational efficiency, even for large network problems, by providing bounds, approximation, or even exact solutions. However, all of the listed methods assume statistical independency between the states of network components and ignore the correlation of the states of network components, e.g., spatial correlations between water pipelines in a seismic disaster situation or the concurrent failure of water generation nodes due to the disconnection of a common power source, which need to be seriously considered in real lifeline network problems. Only a few attempts have been made to fully consider this network component dependency in the non-simulation based methods. Kang et al. [15] estimated the node-pair disconnection probability between each city and a critical facility in a bridge network based on the matrix-based system reliability (MSR) method and considered the correlated failures of bridges by identifying a common source random variable that represents the whole system correlation and makes the conditional independence between the states of components. In this approach, a special path searching algorithm was used to find all of the available paths between the origin and destination nodes. Lim and Song [2] recently proposed a network connectivity analysis method named the selective recursive decomposition algorithm by improving the recursive decomposition algorithm (RDA) developed by Li and He [11]. The selective RDA was applied to lifeline network problems, such as gas and water transmission networks under seismic hazards, and provided quickly converging probability bounds compared to the original RDA by identifying the most reliable disjoint link and cut sets between the origin and destination nodes. The correlations between the failures of the network components were considered in the calculation of each link or cut set by applying the efficient multivariate normal integration method developed by Genz [16]. Although the selective RDA improved the computational efficiency of the original RDA highly, it still needs to deal with the exponentially increasing number of link or cut sets to find the node-pair disconnection probability bounds. Lê and Walter [17] recently developed a probability bounds evaluation method for two-terminal network reliability when the components are statistically dependent. Their methods were applied to various numerical examples and provided reasonable probability bounds, but the tightness of the bounds was highly dependent on the number of components and the pattern of the component correlations.

In this paper, a very fast node-pair connection probability estimation method is proposed for reasonably large-sized networks, especially when the network components are statistically correlated. This method proposes a new recursive procedure for successively compounding the network components from the origin node to the destination node until one single super component representing the connectivity of the origin and destination nodes is obtained. This method uses the sequential compounding method (SCM) [18] to compound the network components, which offers computationally efficient compounding procedures and an extremely low error accumulation. The proposed method does not require an identification of a complex system event, e.g., a consideration of all of the path

connection events, but it automatically considers the system event in the recursive compounding procedure while it fully considers the correlation between the network components. The performance of the proposed method is demonstrated by numerical network examples and a realistic benchmark lifeline network example aiming at a near-real-time reliability estimation of lifeline networks.

2. Proposed method

2.1. Network reliability analysis

In this paper, using the graph theory, a network is defined by a directed acyclic graph $G=(N, L)$ where N and L denote nodes and links in a network, respectively. The network is restricted to an acyclic graph because the path search algorithm and the proposed recursive event identification steps can fall into loops during a path searching process. Nodes are labeled by numbers and links and their directions are represented by the two ordered node numbers at the two ends. In this paper, we use a node weighted modeling by assigning a failure probability to each node and by neglecting the failure probability of the links. Line type elements can be modeled by adding a node in their middle, representing the entire line element with the corresponding failure probability. The reliability of the network is defined by the probability of the connection between an origin node and a destination node, i.e., the probability that the origin node and the destination node are connected through at least one path.

The idea of the proposed method is to compound all of the network components (nodes) until they become a single component to represent the whole network node-pair connection. Fig. 1 illustrates this idea through a simple network example with five nodes. The nodes are indicated by circles, and the links are by directed arrows. In this example, we want to analyse the connectivity between the origin node, node 1, and the destination node, node 4. In step (a), node 5 obviously does not contribute to the connection between nodes 1 and 4; therefore, node 5 and its link from node 2 are removed in step (b). Then nodes 1 and 2 are compounded into super-node A, and nodes 1 and 3 are compounded into super-node B in steps (c) and (d), respectively. These two super-nodes are compounded to super-node C in step (e), and finally, super-node C and node 4 are compounded into super node D. Although this illustration shows the manual compounding procedure schematically, an automated procedure for successive network compounding will be proposed in this paper. In addition, when the failure events of the network components are correlated, compounding is successively performed for two of the components using the sequential compounding methods (SCM) developed by Kang and Song [18]. The details of the compounding procedure offered by SCM for the cases when two components are coupled by intersection and union are summarized in the next section.

2.2. A summary of sequential compounding method

Sequential compounding method (SCM) was developed by Kang and Song [18] to estimate the reliability of a given system composed of correlated components. In SCM, it is assumed that component failures

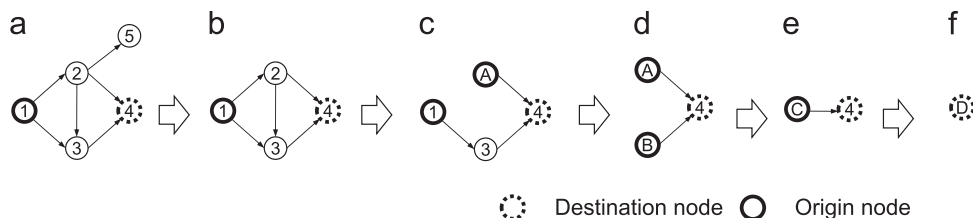


Fig. 1. An illustration of the proposed compounding procedure in a five-node network.

are defined by jointly standard normal variables, e.g., the i th component's failure E_i is defined by $Z_i < \beta_i$ where β_i is the reliability index of E_i . Please note that this assumption can be approximately and more generally applied to non-normal safety margins by linearizing them in transformed normal space using FORM approximation [19]. SCM combines two components coupled by either intersection or union logical operations into a single super component by calculating its probability and the correlations between the super component itself and the other remaining components. This compounding procedure for two components is applied to the whole system in a sequence of pairs until the system becomes a single equivalent component. The procedure for compounding two components coupled by intersection or union is summarized below.

The compounding procedure for two components coupled by *intersection* is as follows. Let us consider two components coupled by intersection whose events are represented by E_1 and E_2 . If we want to combine these two components into a super component represented by the event $E_{1\text{and}2}$, we need to estimate its reliability index (or probability) and the correlation coefficients between the super component itself and the other remaining components. First, the reliability index of the super component is calculated using the following equation:

$$\beta_{1\text{and}2} = -\Phi^{-1}[P(E_1 \cap E_2)] = \Phi^{-1}[\Phi_2(-\beta_1, -\beta_2; \rho_{1,2})] \quad (1)$$

where $\Phi(\cdot)$ is the cumulative distribution function (CDF) of the standard normal distribution, $\Phi_2(\cdot)$ is the joint CDF of the bi-variate standard normal distribution, β_1 and β_2 are the reliability indexes for E_1 and E_2 , and $\rho_{1,2}$ denotes the correlation coefficient between the component events E_1 and E_2 . Second, the correlation coefficient between the super component and each of the other remaining components in the system is calculated as follows. If we consider the correlation between the super component event $E_{1\text{and}2}$ and the k th component event in the network, E_k , $k \neq 1$ or 2 , the correlation coefficient between them can be calculated from the following relationship:

$$\Phi_3(-\beta_1, -\beta_2, -\beta_k; \rho_{1,2}, \rho_{1,k}, \rho_{2,k}) = \Phi_2(-\beta_{1\text{and}2}, -\beta_k; \rho_{(1\text{and}2),k}) \quad (2)$$

where $\Phi_3(\cdot)$ is the joint CDF of the tri-variate standard normal distribution, $\Phi_2(\cdot)$ is the joint CDF of the bi-variate standard normal distribution, β_k is the reliability index of the k th event, and $\rho_{(1\text{and}2),k}$ represents the correlation coefficient between $E_{1\text{and}2}$ and E_k . Since all the terms in Eq. (2) except $\rho_{(1\text{and}2),k}$ are already known, $\rho_{(1\text{and}2),k}$ can easily be solved by minimizing the difference between the left and right hand sides of Eq. (2) using a nonlinear optimization algorithm with the constraint $-1 \leq \rho_{(1\text{and}2),k} \leq 1$. In this paper, we used a MATLAB[®] function called "fminbnd" to obtain the correlation coefficients, which uses the polynomial interpolation and the golden section search [20]. The golden section search technique finds the minimum or maximum of a strictly unimodal function by successively subdividing the range of values according to the existence of the targeted extremum. Although this search does not perfectly guarantee the global extremum and relies on the selection of a starting point for a general problem, it provides an accurate estimation of extremum for this kind of approximation when estimating a correlation coefficient ranging from -1 to 1 . The use of an advanced optimization algorithm will improve the accuracy and/or efficiency of the correlation coefficient estimation in the proposed method.

To solve Eq. (2) more efficiently, by conditioning the left and right-hand sides of Eq. (2) by the commonly included component k , we can use the following approximated equation instead [18]:

$$\Phi_2(-\beta_{1|k}, -\beta_{2|k}; \rho_{1,2|k}) = \Phi(-\beta_{(1\text{and}2)|k}) \quad (3a)$$

All terms in Eq. (3a) are determined exactly as [21]:

$$\beta_{1|k} = (\beta_1 - \rho_{1,k}A_k) / \sqrt{1 - \rho_{1,k}^2 B_k}$$

$$\begin{aligned} \beta_{2|k} &= (\beta_2 - \rho_{2,k}A_k) / \sqrt{1 - \rho_{2,k}^2 B_k} \\ \rho_{1,2|k} &= (\rho_{1,2} - \rho_{1,k}\rho_{2,k}B_k) / \left(\sqrt{1 - \rho_{1,k}^2 B_k} \sqrt{1 - \rho_{2,k}^2 B_k} \right) \\ \beta_{(1\text{and}2)|k} &= (\beta_{1\text{and}2} - \rho_{(1\text{and}2),k}A_k) / \sqrt{1 - \rho_{(1\text{and}2),k}^2 B_k} \end{aligned} \quad (3b)$$

where $A_k = \varphi(-\beta_k)/\Phi(-\beta_k)$; and $B_k = A(-\beta_k + A)$ in which $\varphi(\cdot)$ denotes the PDF of the standard normal distribution. Eq. (3a) is an approximation because it is assumed that the conditional probabilities are calculated using normal distributions; however, they may follow non-normal distributions [22].

The compounding procedure for two components coupled by *union* is as follows. Let us consider again the two components whose events are represented by E_1 and E_2 , but in this case these two components are coupled by *union*. If we combine these two components into a super component represented by the event $E_{1\text{or}2}$, the reliability index of the super component and the correlation coefficients between the super component and the other remaining components are estimated as follows. First, the reliability index of the super component is calculated by the following equation by slightly modifying Eq. (1) using De Morgan's law:

$$\beta_{1\text{or}2} = -\Phi^{-1}[P(E_1 \cup E_2)] = \Phi^{-1}[P(\bar{E}_1 \cap \bar{E}_2)] = \Phi^{-1}[\Phi_2(\beta_1, \beta_2; \rho_{1,2})] \quad (4)$$

Second, the correlation coefficient between the super component and each of the other remaining components represented by E_k , $k \neq 1$ or 2 , in the system is calculated from the following relationship:

$$\Phi_3(\beta_1, \beta_2, \beta_k; \rho_{1,2}, \rho_{1,k}, \rho_{2,k}) = \Phi_2(\beta_{1\text{or}2}, \beta_k; \rho_{(1\text{or}2),k}) \quad (5)$$

where the only unknown term $\rho_{(1\text{or}2),k}$ is solved by minimizing the difference between the left and right hand sides of Eq. (5) using a nonlinear optimization algorithm with the constraint $-1 \leq \rho_{(1\text{and}2),k} \leq 1$. To obtain better computational efficiency, the same as in the intersection case, Eq. (5) is approximated as

$$1 - \Phi_2(\beta_{1|k}, \beta_{2|k}; \rho_{1,2|k}) = \Phi(-\beta_{(1\text{or}2)|k}) \quad (6a)$$

where

$$\beta_{(1\text{or}2)|k} = (\beta_{1\text{or}2} - \rho_{(1\text{or}2),k}A_k) / \sqrt{1 - \rho_{(1\text{or}2),k}^2 B_k} \quad (6b)$$

SCM's accuracy and efficiency for series, parallel, cut-set, and link-set systems has been demonstrated through various numerical examples with up to 1000 components, and it has been confirmed that the accuracy of the method is not significantly compromised by a large number of components. However, SCM cannot be used in a network connectivity analysis directly and still needs to be further developed due to the following two reasons: first, SCM can only be applied to a system reliability estimation problem when a system event is already given in a logical function form. However, in a network reliability analysis, a system event identification is not a simple task because the identification of a node-pair connectivity event requires finding all of the available paths between the origin and destination nodes using a special path search algorithm. For large-sized networks, this is almost impracticable. Second, the practical application of SCM is limited to the compounding of all of the different components. If the identical event appears multiple times in a system event, those events should be duplicated by assigning the same reliability index and the same correlation coefficients. For example, in the system event defined by three components, such as $E_{\text{sys}} = E_1 E_2 \cup E_1 E_3 \cup E_2 E_3$, if we combine E_1 and E_2 into a super component with event E_A using Eq. (1) and (3), we also need to estimate the correlation coefficient between E_A and E_1 where E_A already includes E_1 . In this case, we have to duplicate E_1 by assigning the same reliability index and the same correlation coefficients to use E_1 twice. If such an identical event appears many times, it increases the size of the problem. In a network node-pair connectivity event, this

duplication is repeated many times as there are many components commonly included in multiple paths connecting the origin and destination nodes, and this issue increases the problem size and the computational cost. For these reasons, SCM cannot be used in a node-pair connectivity analysis directly. To overcome these issues, the following section proposes a new approach for network reliability analysis using the compounding procedures in SCM.

2.3. The proposed method

In this section, a new recursive algorithm is proposed based on SCM to compound the components in a network from the origin node to the destination node. The main idea of the compounding procedure is to compound all of the network nodes from the origin node to the destination node to represent the connectivity between the origin and destination nodes. As a schematic illustration, we consider the simple network example in Fig. 1, which has five nodes and six links. Nodes 1 and 4 in this network are assigned as origin and destination nodes, respectively. We herein estimate the probability of connection between those two nodes. We assign events of reliability to the five nodes as E_1, E_2, E_3, E_4 , and E_5 , each of which represents the reliable state of each node. The compounding process of this network takes place as follows: first, we remove node 5 as shown in Fig. 1 (b) because it does not contribute to the connection event between nodes 1 and 4. Second, we combine nodes 1 and 2 into a super node A as shown in Fig. 1 (c), and combine nodes 1 and 3 into a super node B as shown in Fig. 1 (d), where the event of the super nodes A and B are defined by $E_A = E_1 E_2$ and $E_B = E_1 E_3$ and each super component is defined by the intersection of the two node events to represent the connection between the two nodes. In these compounding processes, since node 1 is connected to and affects the connectivity of both nodes 2 and 3, it is duplicated and compounded twice. Then, the link between nodes 2 and 3 is removed, because it is a redundant link after the compounding process. Third, because this network is reliable if either node A or B is connected to node 4, we combine nodes A and B with node 4 and make a super component D whose event is defined by $E_D = (E_A \cup E_B) E_4$. Finally, the network reliability is equal to the reliability of the super component D. For verification purposes, we can manually construct the system event of the connection between nodes 1 and 4 by identifying all of the paths between these two nodes, i.e., {1, 2, 4}, {1, 3, 4}, and {1, 2, 3, 4}. Then, the corresponding system event is defined as

$$E_{sys} = E_1 E_2 E_4 \cup E_1 E_3 E_4 \cup E_1 E_2 E_3 E_4 = (E_2 \cup E_3) E_1 E_4 \quad (7)$$

If we identify the system event using the proposed scheme, the system event is

$$E_{sys} = E_D = (E_A \cup E_B) E_4 = (E_1 E_2 \cup E_1 E_3) E_4 = (E_2 \cup E_3) E_1 E_4 \quad (8)$$

From Eqs. (7) and (8), we can confirm that the identical system events are derived. For the probability calculation of Eq. (8), the probabilities of E_A, E_B , and E_D can be calculated simply by using the compounding processes in the previous section.

Since this example is a small network, we can easily compound all the network components manually. However, for a node-pair connectivity analysis of a large general network, manual compounding of the components is impractical. Therefore, this paper proposes a recursive algorithm for the automatic compounding of components in a general network, which consists of the steps 0–4 below. This automated procedures will be illustrated using the small network example in Fig. 1.

Step 0: remove unnecessary nodes.

(0-1) Remove all incoming links to the origin node (not outgoing links from the origin node), and all outgoing links

from the destination node (not incoming links to the destination node). (0-2) Remove all nodes that have no incoming links or all nodes that have no outgoing links (including isolated degree-0 nodes), except the origin and destination nodes, because they do not contribute to the system event. In (0-1) and (0-2), remove any links connected to these nodes as well.

Repeat (0-1) and (0-2) until there are no such end nodes except the origin and destination nodes.

Step 1: recursively search for a node that is connected only by the origin node(s).

This step is a backward recursive searching process from the destination node to the origin node.

Check all incoming neighbors of the destination node (all adjacent nodes having links to the destination node).

If the only incoming neighbor node is the origin node, go to step 2 and start compounding. If not, recursively check each of the incoming neighbor nodes if its incoming nodes are only the origin node(s). If such a node is found, go to step 2. Note that multiple origin nodes can be generated in steps 2–4.

Step 2: compound origin node(s).

If more than one origin nodes are connected to the node identified in step 1, combine all of the connected origin nodes into a super component using the compounding procedure in Eq. (6), as they are coupled by *union*. Otherwise, go to step 3. From this compounding process, the reliability index of the super node and the correlation between this super component and the other remaining components will be calculated.

Additional rule for step 2 if any of the origin nodes have outgoing links towards nodes other than the node identified in step 1, those nodes need to be duplicated before the compounding process. Duplication means making one more node identical to the origin node, with the same label, the same reliability index, and the same correlation coefficients. The purpose of the duplication is to keep those nodes after the compounding process for further compounding with the other connected nodes.

Step 3: compound origin node and the node identified in step 1.

Now, only one origin node is connected to the node identified in step 1. Combine the one origin node and the node identified in step 1 into a super component using Eq. (3), as they are coupled by *intersection*. From this compounding process, the reliability index and the correlation coefficients of the super node are estimated. This new super node becomes a new origin node.

Additional rule for step 3 after the compounding in step 3, check if the super node and duplicated origin nodes in the *Additional rule for step 2* have common links to certain nodes. If so, remove those links for computational efficiency as they are just redundant links in the system event.

Step 4: recursively repeat steps 1–3 until obtaining a single super component.

After step 3, go back to the previous step of the recursion, and repeat steps 1–3 until compounding all the nodes in the given network. Finally, the destination node becomes a single super node that represents the connectivity between the origin and destination nodes. The reliability of the network can be calculated simply as $\Phi(-\beta_{supermode})$, where $\Phi(\cdot)$ denotes the cumulative distribution function of the standard normal distribution and $\beta_{supermode}$ denotes the reliability index of the single super node.

This proposed network component compounding algorithm can be applied to any generic directed acyclic networks. This algorithm is also illustrated as a flow chart in Fig. 2.

The proposed algorithm has computational cost benefits over the direct use of SCM for network problems due to the following

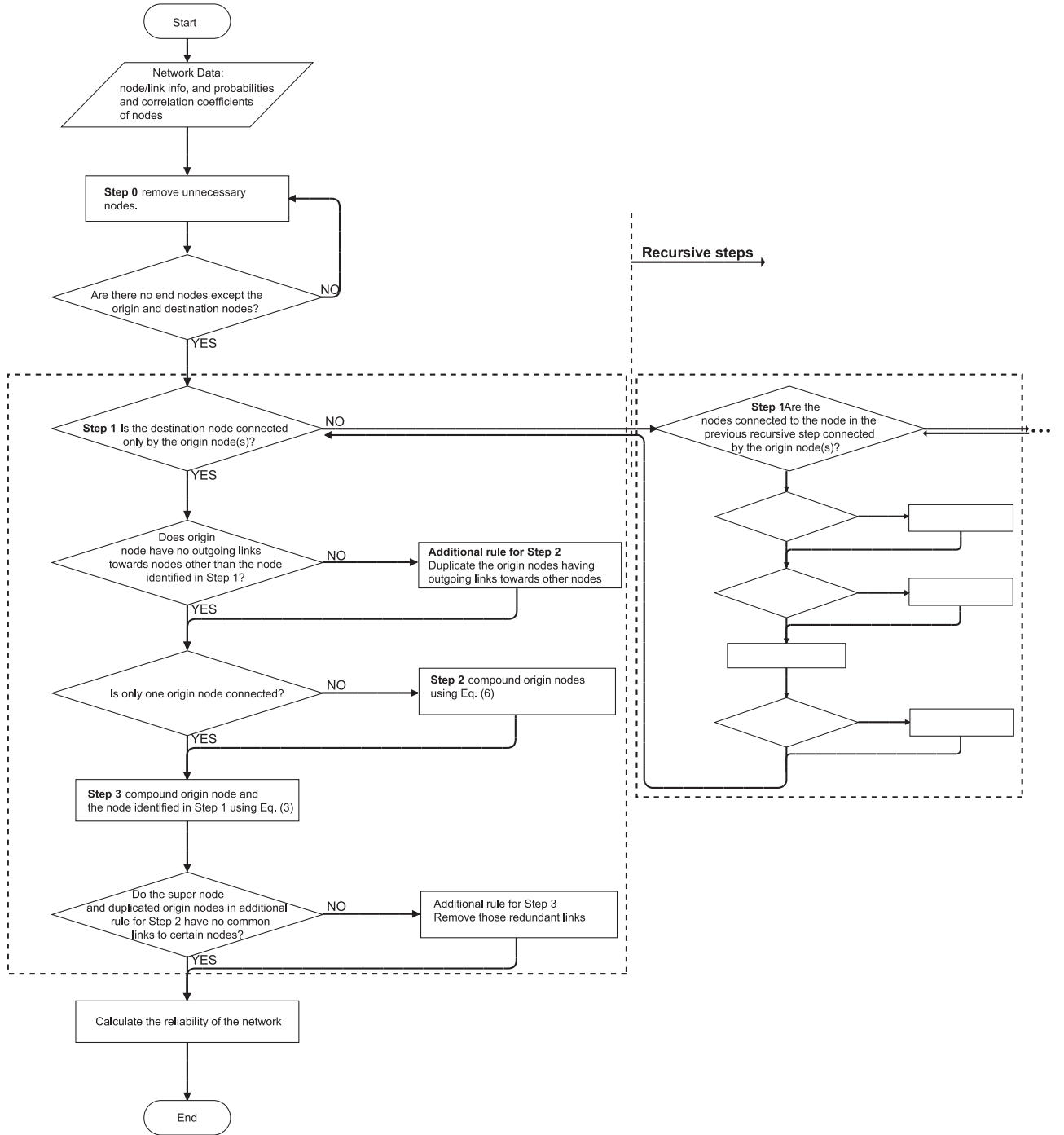


Fig. 2. A flow chart of the proposed recursive algorithm.

reasons: (1) the compounding procedure considers all link set events automatically instead of identifying all paths between the origin and destination nodes, and (2) the size of the correlation matrix decreases as the compounding procedures are carried out, or at least remains if a redundant node is added in step 3. The matrix size keeps decreasing during the compounding procedure in steps 3 and 4.

As an illustrative example, the connection probability of the network given in Fig. 1 is estimated using the proposed recursive algorithm. First, we assume that the reliability of each of the five nodes is 95% equally (or the reliability index $\beta_i = -1.26$ for $i=1, \dots, 5$). The correlation coefficient between the i th and j th

nodes ($i=1, \dots, 5, j=1, \dots, 5, i \neq j$) are all assumed to be identical at $\rho_{ij}=0.2$. The corresponding correlation matrix is

$$\mathbf{R} = \begin{bmatrix} 1 & 0.2 & 0.2 & 0.2 & 0.2 \\ & 1 & 0.2 & 0.2 & 0.2 \\ & & 1 & 0.2 & 0.2 \\ & & \text{sym.} & 1 & 0.2 \\ & & & & 1 \end{bmatrix} \quad (9)$$

In this example, for demonstration purposes, each compounding step is shown along with the updated probability and correlation

coefficients. First, according to *step 0*, we remove the unnecessary node 5, which has no outgoing links. Second, according to *step 1*, we check the destination node (node 4) to see if it is connected only by the origin node (node 1). Node 4 is connected by nodes 2 and 3 but these nodes are not the origin node, and therefore, we repeat *step 1* recursively for each of nodes 2 and 3 to check if they are connected only by the origin node. Then, we find that node 2 is connected only by the origin node (node 1). Since node 1 is the only single origin node connected to node 2, we skip *step 2*. In *step 3*, we combine nodes 1 and 2 into node A as shown in Fig. 1 (c). The reliability of node A is calculated using Eq. (1) as $P_A=0.817$, and the correlation matrix of the nodes 1, A, 3, and 4 is estimated using Eq. (3) as

$$\mathbf{R} = \begin{bmatrix} 1 & \mathbf{1} & 0.2 & 0.2 \\ & 1 & \mathbf{0.226} & \mathbf{0.226} \\ & \text{sym.} & 1 & 0.2 \\ & & & 1 \end{bmatrix} \quad (10)$$

where the updated values are shown in bold. Node 1 still remains because of the duplication in the *Additional rule for step 2* to calculate the correlation coefficient between nodes 1 and A. In addition, according to the *Additional rule for step 3*, the redundant links between nodes 2 and 3 are removed. Now, node A becomes a new origin node, and there are a total of two origin nodes, node 1 and node A. According to *step 4*, we go back to the previous step of the recursive algorithm and check node 3. Now, node 3 is connected only by an origin node, node 1. Since no more origin node is connected to node 3, we skip *step 2* and follow *step 3* to compound nodes 1 and 3 into a super node, node B as shown in Fig. 1 (d). Then, node B becomes a new origin node, and there are a total of two origin nodes, nodes A and B. The reliability of node B is calculated using Eq. (1) as $P_B=0.817$, which is the same as the reliability of node A, and the correlation matrix of nodes A, B, and 4 is estimated using Eq. (3) as

$$\mathbf{R} = \begin{bmatrix} 1 & \mathbf{0.773} & 0.226 \\ & 1 & 0.226 \\ & \text{sym.} & 1 \end{bmatrix} \quad (11)$$

where the updated value is shown in bold. In *step 4*, we go back to the previous recursive step where we check if node 4 is connected only by the origin nodes. In this case, node 4 is connected by nodes A and B which are all origin nodes. Therefore, according to *step 2*, we compound nodes A and B into a super node, node C. Here, Eq. (4) and (6) are used for the compounding process for two components coupled by union. Finally, according to *step 3*, nodes C and 4 are compounded using Eq. (1) and (3), and we have only one super node, node D, which represents the connectivity between nodes 1 and 4. The reliability of node D is $P_D=0.809$, and it is cross-checked by the Monte Carlo Simulations with 10^6 samplings where the reliability is obtained as 0.808.

2.4. Error accumulation and improvement of the proposed method using a heuristic approach

Although Kang and Song [18] demonstrated that SCM's error accumulation is almost negligible even for a very large system with 1000 different correlated components, SCM still has a possibility of large error accumulation due to the approximation nature of its assumptions which are not fully mathematically proven. One of the assumptions that needs to be carefully considered is that the distribution of super components compounded from Eqs. (1), (3), (4), and (6) still follows the normal distribution, and the joint distribution between these super components follows the multivariate normal distribution. Although this assumption does not accumulate significant errors in most cases as demonstrated by a variety of numerical examples in Kang and Song [18], an exceptional case is found when compounding two or more super components that include common components. To test the error accumulation of this assumption, we test a bench mark problem with the link set event:

$$E_{\text{sys}} = E_1 E_2 \cup E_1 E_3 \cup E_1 E_4 \cup \dots \cup E_1 E_i \cup \dots \cup E_1 E_n \quad (12)$$

where $E_1, E_2, \dots, E_n$ represent the events of n components. The probability of Eq. (12) can be estimated using SCM in two different ways. The first way is to calculate the probability of Eq. (12) by

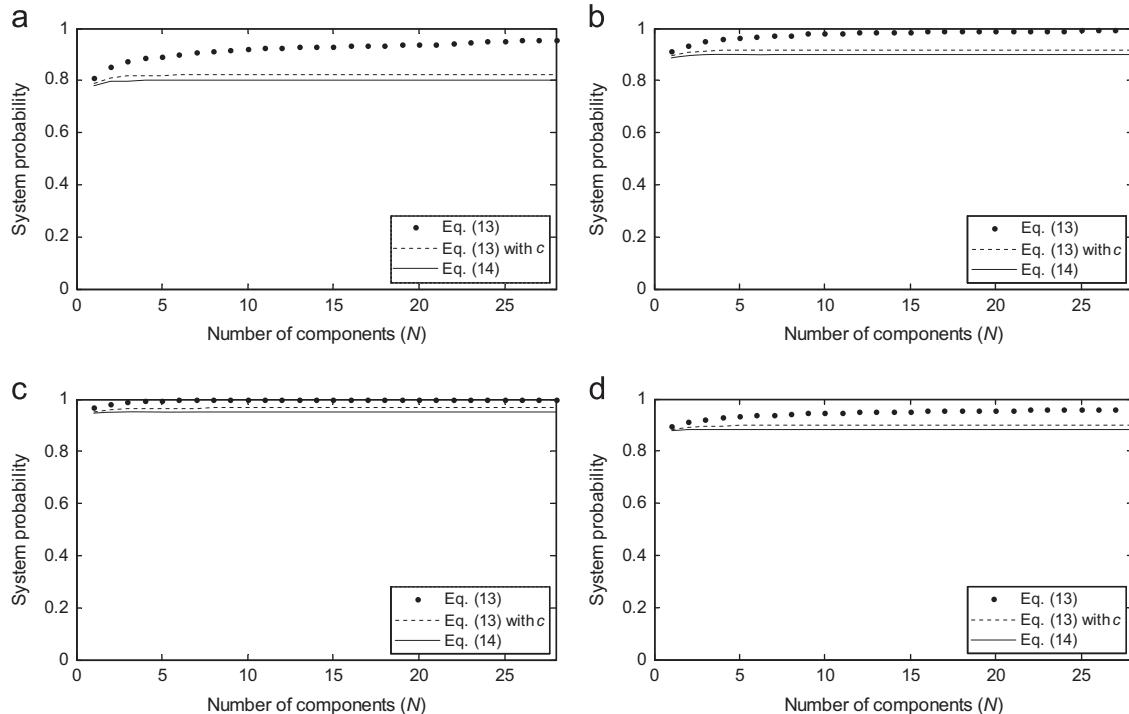


Fig. 3. Result comparison between Eqs. (13) and (14). (a) $P(E_i)=0.8$, random correlation; (b) $P(E_i)=0.9$, random correlation; (c) $P(E_i)=0.95$, random correlation; and (d) $P(E_i)=0.90$, equi-correlation with $\rho=0.8$.

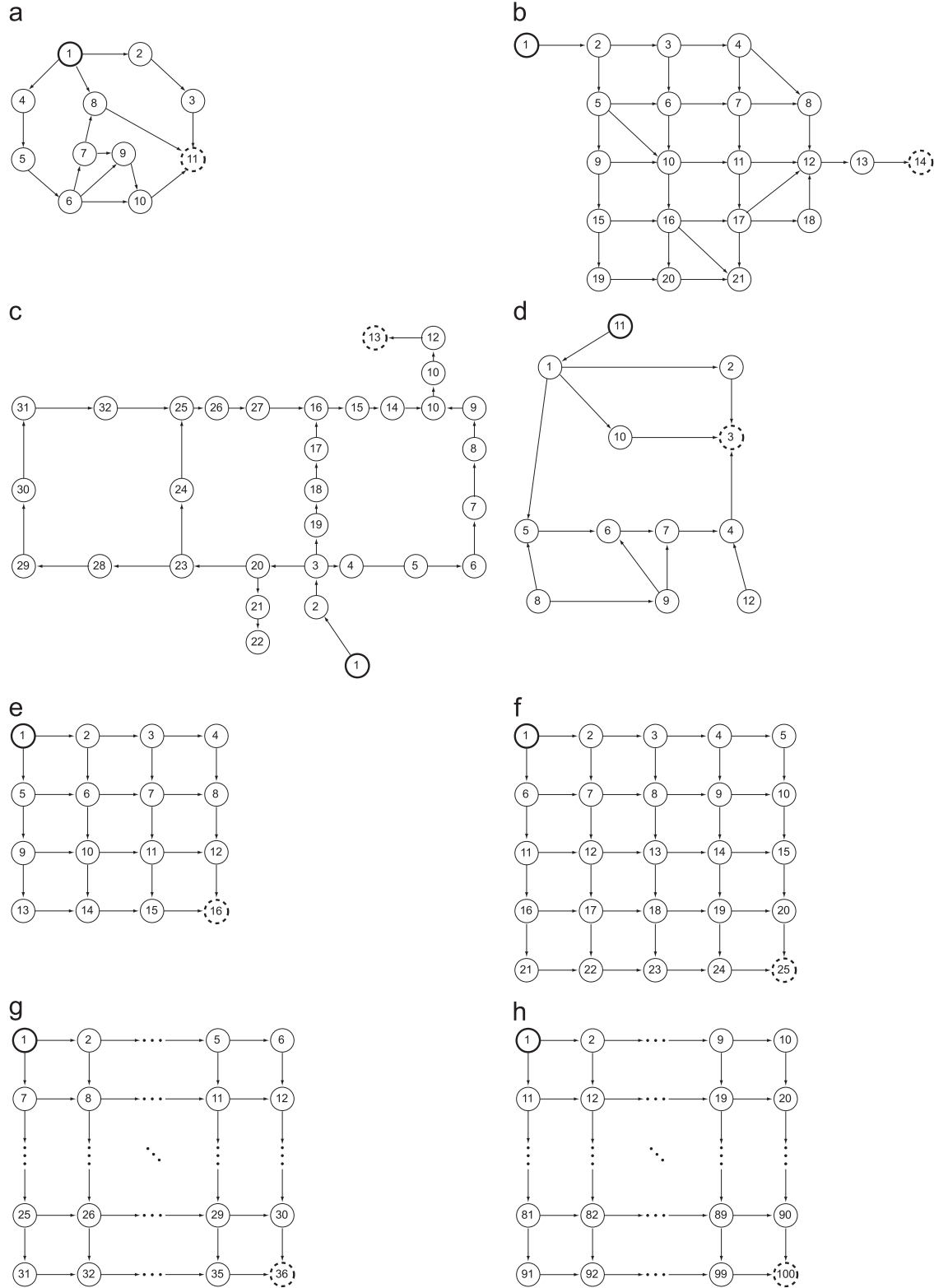


Fig. 4. Four numerical network examples. (a) 11 nodes/15 links, (b) 21 nodes/34 links, (c) 32 nodes/34 links, (d) 12 nodes/15 links, (e) 16 nodes/24 links, (f) 25 nodes/40 links, (g) 36 nodes/60 links, and (h) 100 nodes/180 links.

compounding E_1 and E_i into a super component first, and then compound all of the super components into a single super component as follows:

$$P(E_{\text{sys}}) = P(E_{S1} \cup E_{S2} \cup \dots \cup E_{Si} \cup \dots \cup E_{Sn}) \quad (13)$$

where $E_{Si} = E_1 E_i$. In this case, all of the super components include E_1 , which is the common component. The second way is to calculate Eq. (12) by compounding $E_2 - E_n$ into a single super component E_S , and then compound this super component with E_1 as follows:

$$P(E_{\text{sys}}) = P(E_1 E_S) \quad (14)$$

where $E_S = E_2 \cup E_3 \cup \dots \cup E_i \cup \dots \cup E_n$. Although Eqs. (13) and (14) are identical, Eq. (14) can avoid the common component between the super components. To test these two equations, the probability results of both equations are calculated for the various values of the components, component reliabilities, and the correlations between the components, and the representative cases are plotted in Fig. 3. For (a)–(c), the component reliability is assumed to be identical, and the values are given as (a) $P(E_i)=0.8$, (b) $P(E_i)=0.9$, and (c) $P(E_i)=0.95$. The correlation coefficients between the components are randomly generated to have values between 0 and 1 using the algorithm developed by Numpacharoen and Atsawarungruangkit [23]. For (d), $P(E_i)=0.95$ is assigned to all of the components equally, and the correlation coefficient is assumed to be identical at 0.8. Contrary to the expectation that the results of both equations should be identical because they are the same equations, Eq. (13) always overestimates the system probability when comparing to Eq. (14). The reason for Eq. (13)'s overestimation error is that the super components involve the common component E_1 , which adds nonlinearity to the limit state functions of the super components and makes the multivariate normal distribution assumption between the super components invalid. Once the common component E_1 is compounded into super components, the compounding between those super components no longer include this common component in the rest of the compounding processes. In other words, the contribution of the common component to the correlation between the super components is neglected, and it decreases the correlation coefficient between super components, resulting in the overestimation in the probability calculation of Eq. (13), especially when calculating the union of the events. Obviously, it is not a simple task to find the exact multivariate distribution type for these super components, and therefore, we propose a heuristic approach to significantly reduce these overestimation errors while keeping the multivariate normal assumption. In Fig. 3, we can observe that the estimation errors are always overestimations, and they keep increasing with the size of the system. Therefore, we propose a correction factor c , which is multiplied to the probability of the common component E_1 whenever this common component is duplicated. For example, if E_1 appears twice, the second E_1 , the duplicated event, has the probability $P(E_1) \times c$. If this duplicated event is duplicated again, the third identical event has the probability $P(E_1) \times c \times c$. The j th duplicated event has the probability $P(E_1) \times c^j$. The effect of this correction factor is to reduce the probability of the common component and to result in the reduction of the system probability. The value of c is proposed to be 0.93 in this paper, which is obtained by minimizing the difference between the results of Eqs. (13) and (14) for many random values of component reliability, correlation coefficients, and the number of components. The best-fit values of the correction factor c ranges between 0.90 and 0.95, and the average value 0.93 is suggested. Since this correction factor c is obtained based on Eqs. (13) and (14), the value 0.93 is valid only for link set formulations such as network

events, and it will have a different value for different system event definitions. This heuristic approach approximately estimates the non-normal multivariate distribution by simply considering the reduction of the probability of the duplicated events and does not consider the effect of the correlation coefficients for simple implementation. The performance of this approach is also demonstrated in Fig. 3, where the proposed approach always reduces the overestimation of Eq. (13) significantly. The performance of this heuristic approach will also be demonstrated by the numerical network examples and realistic interconnected networks in the following sections. Although this approach is just a simple modification, it gives very effective error reduction and it prevents error accumulation for a large number of components. Also, this duplication procedure is not always carried out during each step of the compounding process of the network components. However, this simple improvement still has a limitation, as it cannot be mathematically proven and its error cannot be exactly estimated in advance. In this paper, in all of the analyses, this heuristic approach is applied when a network component is duplicated according to the *Additional rule for step 2* in the previous section.

3. Numerical examples for connectivity analysis

In this section, the method proposed in this paper is applied to the eight numerical network examples shown in Fig. 4(a)–(h). The first four examples (a)–(d) were developed based on the topology of the network examples in the literature: (a) Yang et al. [24], (b) Gupta et al. [25], (c) Fujiwara and Kang [26], Soltanijalili et al. [27], and (d) Sevük and Altinbilek [28], and the other four examples (e)–(h) were developed as ideal grid networks. The node failure probability and correlation coefficients were arbitrarily assigned for all of these network examples. Network (a) consists of 11 nodes and 15 links, network (b) consists of 21 nodes and 34 links, network (c) consists of 32 nodes and 34 links, network (d) consists of 12 nodes and 15 links, network (e) consists of 16 nodes and 24 links, network (f) consists of 25 nodes and 40 links, network (g) consists of 36 nodes and 60 links, and network (h) consists of 100 nodes and 180 links. All of these examples are represented by directed acyclic graphs, and their origin and destination nodes are indicated by a bold circle and a dotted bold circle, respectively. These examples were developed for demonstration purposes, and it is assumed that the reliabilities of all of the nodes are identical and all of the nodes are equi-correlated in each example for simplicity. In Table 1, the reliability of each node and the correlation coefficients between the nodes are presented. Only node failures are considered; link failures are not considered for simplicity. For these eight network examples, connectivity analyses between the origin and destination nodes were performed using the proposed method, and the reliability and computation times are reported in Table 1. For these analyses, MATLAB[®] was used on a computer with an Intel I7 CPU (2.80 GHz each) and 3 GB of RAM. The results were confirmed by the Monte

Table 1
Connectivity analysis results for the four examples in Fig. 1.

Example	Reliability of each node	Correlation coefficient	Proposed method		MCS with 100,000 samples	
			Connection probability	Computation time (s)	Connection probability	Computation time (s)
(a)	0.8	0.2	0.633	0.587	0.628	43.702
(b)	0.995	0.3	0.973	2.148	0.976	99.025
(c)	0.95	0.7	0.823	4.131	0.827	88.361
(d)	0.85	0.5	0.698	0.272	0.685	43.432
(e)	0.4	0.4	0.122	1.273	0.135	25.633
(f)	0.9	0.7	0.813	3.378	0.817	86.738
(g)	0.99	0.6	0.961	7.867	0.979	136.468
(h)	0.95	0.5	0.904	65.598	0.890	414.199

Carlo simulations (MCS) done on the same computer. The results demonstrate that the proposed method shows reasonable accuracy in comparison with MCS with 100,000 samplings for all eight of the network problems, and the computational efficiency is significantly high. For MCS, 100,000 samplings were chosen as it gives sufficient convergence in the results. The computation time for MCS is also reported in Table 1, where the “mvnrnd” MATLAB[®] function was used to quickly generate the multivariate normal random numbers. This MCS computation time is reported only for reference purposes, not for comparison purposes, because the proposed method and MCS cannot be compared fairly due to their different natures. For example, MCS has a sampling nature, and its computation time depends on both the magnitude of the probability and the network size, while the proposed method's computation time just depends on the network size. In addition, the MCS error is measured by its convergence rate or confidence bounds, but the proposed method's error is impossible to be measured in advance.

The computational efficiency of the proposed method depends primarily on the number of links between the origin and destination nodes. This is because the method's compounding process consumes most of the computational costs when the correlation between the compound component and the other remaining components are numerically estimated, and the cost is proportional to the number of the remaining components/links. As the compounding process continues, the number of remaining links is gradually reduced, as the links are merged into compounded components. Therefore, the computational requirement can be represented by $O(N_L^2)$ where N_L denotes the number of links. This is clearly illustrated in the same type of grid network examples (e)–(h) in Fig. 4 in which the computational time is directly proportional to the square of the number of links.

4. Application to interdependent networks

In this section, the proposed algorithm is applied to a benchmark lifeline network example, interdependent power and water networks in Memphis/Shelby County, as shown in Fig. 5. Fig. 5(a) shows a power grid consisting of 8 power generation nodes and 51 power distribution nodes, and Fig. 5(b) shows a water network in the same area consisting of 15 water generation nodes and 34 water distribution nodes. It is assumed that the 15 water generation nodes are supplied by 15 out of the 51 power distribution nodes [29]. A backup supply node is introduced to each water generation node to model the interdependency level between the power and water networks. The interdependency level is defined by the failure probability of each backup supply node, which has a value between 0 (independency) and 1 (100% dependency) [29]. The interdependency level in this example is assumed to be equally at 0.5. A set of earthquake events with an epicenter at Marked Tree, Arkansas (35.535N, 90.430W) is considered, and the ground motion parameters at the location of each node are estimated using the Mid-America Earthquake Center's loss estimation software (MAEViz) [30]. It is assumed that all links have one-directional flows, and there is no loop, based on the previous study [29] to avoid unrealistic counter flows. For simplicity, only node failures are considered and pipe line segment failures are not considered. The failure probability of each node is calculated from the fragility curves in HAZUS-MH [31] by assigning the groups of structures and fragility curve parameters according to Kim et al. [32]. As there is no specific correlation model for the Shelby County area, we assume that the correlation coefficients between these nodes follow the macro-spatial auto-correlation model proposed by Wang and Takada [33] using the correlation length of the Chi–Chi earthquake. The values of the correlation coefficients obtained from this auto-correlation model range between 0.130 and

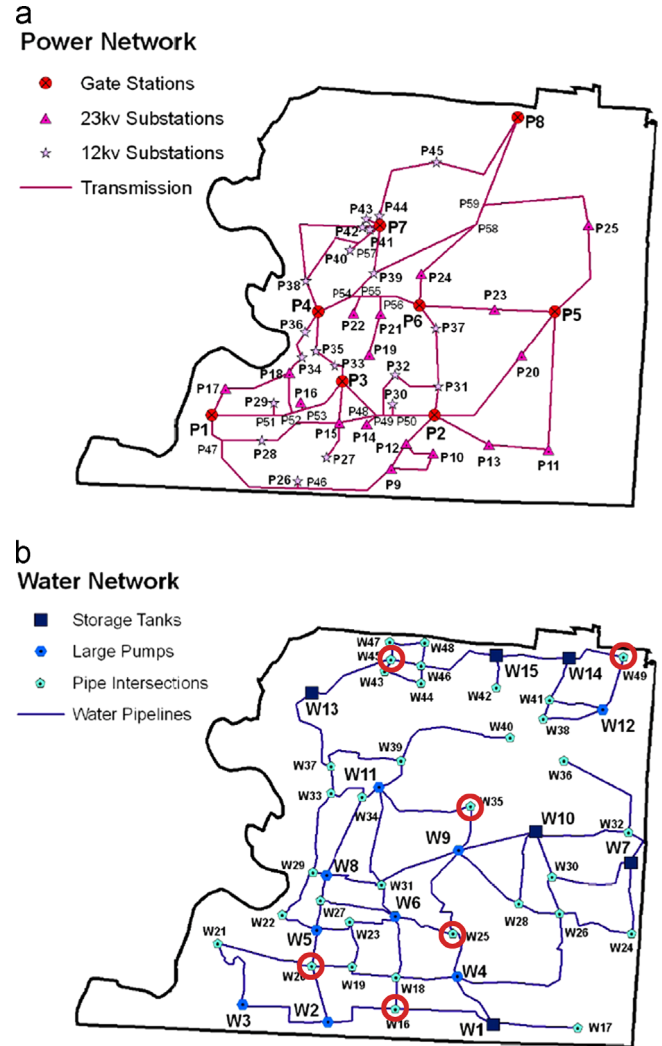


Fig. 5. Memphis/Shelby County power and water networks [34].

0.996 according to the proximity between nodes. In this example, the connectivity is defined by the system event in which at least one power source, i.e., a power generation node or a backup supply node, is connected to a water distribution node. The condition that at least one power source is connected is modeled by introducing an additional virtual node, which connects all of the power generation nodes and backup supply nodes. It is assumed that the virtual node is 100% reliable, and that it is statistically independent of all the other network nodes.

The connectivity analysis for this system event is performed using the proposed algorithm in this paper, and it is repeated for a range of earthquake magnitudes between 5.5 and 8.2. The results of the connectivity analysis are plotted in Fig. 6 for six connection scenarios to the water distribution nodes, WD16, WD20, WD25, WD35, WD45, and WD49 (circled in Fig. 5) as representative cases. These plots show the different seismic risk levels of the water distribution nodes in terms of their functionality. To verify these results, the same analyses were performed using MCS with 100,000 samples and compared with the results of the proposed method in the same figure. Overall, the results from the proposed method and MCS show good agreement with each other in all six of the plots. The worst case in terms of accuracy is found in Fig. 6(a), and the maximum relative error is observed as 31.91%, where the relative error is calculated in terms of disconnection probability instead of connection probability to show the error clearly even for high reliability cases. In all the other cases,

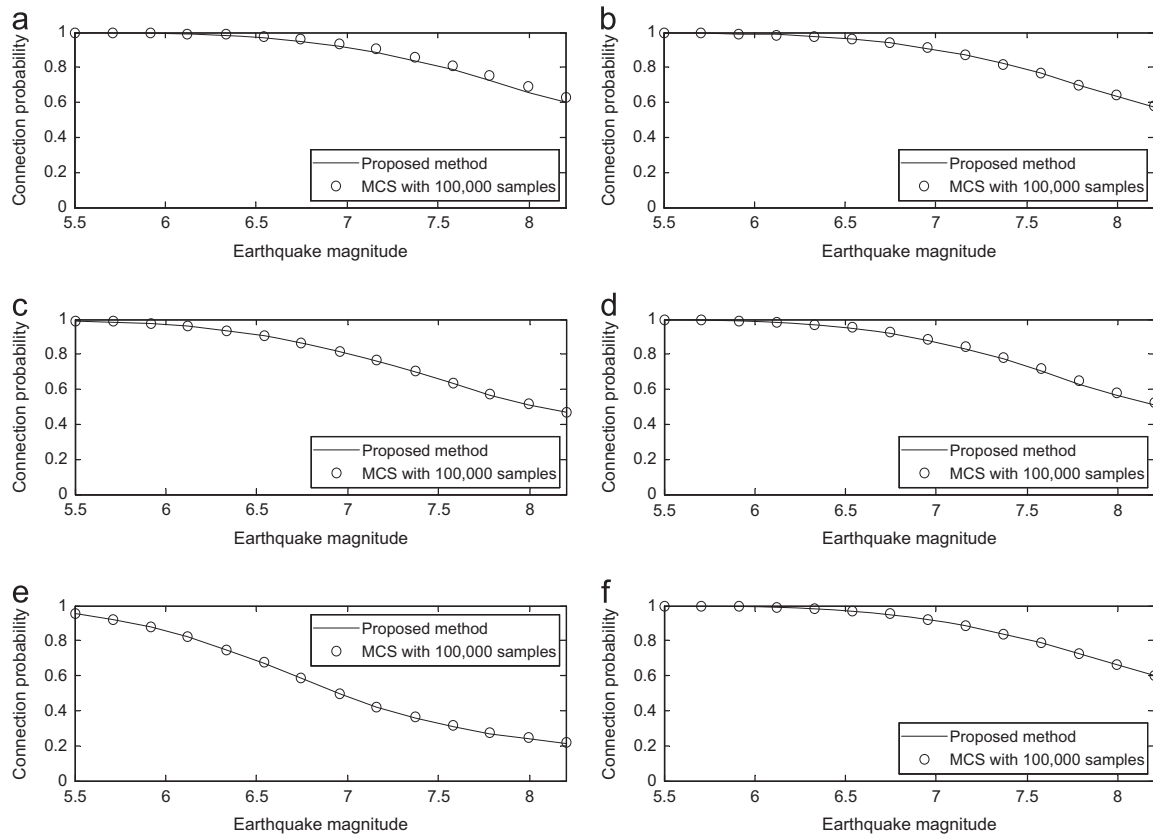


Fig. 6. Connectivity analysis results of the interconnected power and water networks. (a) WD16, (b) WD20, (c) WD25, (d) WD35, (e) WD45, and (f) WD49.

less or even negligible error accumulation is observed. This error is accumulated from the assumption that the distribution type of the super components does not follow the multivariate normal distribution while no significant error arises from the calculations in Eqs. (1), (3), (4), and (6). The computation time of each analysis of each plot is reported in Table 2, where each plot consists of 14 analyses. The proposed method again shows remarkable computational efficiency, even for a large lifeline network example with statistically correlated components. Since the reported computation time is based on a single analysis, much more computational cost saving is expected for multiple analyses such as making the whole plots in Fig. 6. Computational efficiency in network reliability analysis is particularly important when urgent post-hazard decision making is needed or considerations of different disaster scenarios with repeated risk assessments are required. The MCS computational time is also listed in Table 2 as reference values, but again, this is not for comparison purposes, because there is no way to fairly compare the proposed method and MCS due to their different analytical and statistical natures. The computation time was recorded based on MATLAB[®] on a computer with an Intel I7 CPU (2.80 GHz each) and 3 GB of RAM. The proposed method's high computational efficiency can be obtained due to the exclusion of the non-contributing network components to the connectivity events, the fast compounding procedures, the non-increasing correlation matrix size due to the compounding nature, and low error accumulation due to SCM and the proposed heuristic improvement. However, this result does not guarantee the high efficiency of the proposed method for extremely large-sized networks with more than 1000 components, as its computational efficiency will be highly affected by the network size and the correlation matrix size. An additional limitation of this method is that there is no way to measure the exact error or error bounds of the analysis results, as any error arising from the joint normal assumptions of super components and the proposed heuristic approach is impossible to measure in advance.

Table 2

Computational cost for each analysis in Fig. 6.

Case	Destination node	Computation time per analysis on average (s)	
		Proposed method	MCS with 100,000 samples
(a)	WD16	1.058	674.317
(b)	WD20	1.055	503.648
(c)	WD25	3.449	874.703
(d)	WD35	1.341	695.320
(e)	WD45	0.218	617.136
(f)	WD49	0.432	505.311

5. Conclusion

A non-simulation-based reliability analysis method was developed for node-pair connectivity analysis of lifeline networks under seismic hazards. The main advantage of the proposed method is that it rapidly estimates a node-pair connection probability of general directed acyclic networks especially when the network components are statistically correlated. The sequential compounding method (SCM), which was originally developed for the reliability analysis of correlated general system events, was integrated into a bottom-up recursive compounding process to compound all of the network components into a single super component. A heuristic approach was proposed to reduce the error accumulation from the compounding procedures of super components that share common network components to improve the accuracy of the proposed method. The proposed method was applied to numerical network examples and interconnected power and water networks in Shelby County, USA, as benchmark examples to estimate the reliability of these networks in terms of the connectivity of the origin and destination nodes. The accuracy of the proposed method was comparable with that of the

Monte Carlo Simulations with 100,000 samplings throughout these examples, and the error accumulation was not significant in most cases. The computational efficiency of the proposed method was remarkable, even for the realistic interdependent lifeline network example as demonstrated through six different connectivity scenarios. The advantages and possible limitations of the proposed method in terms of computational cost were addressed.

Acknowledgement

The first author gratefully acknowledges that this work is supported by the University of Western Sydney under the UWS Research Seed Grant Scheme, the UWS International Research Initiatives Scheme, and the IIE Research Grant Scheme.

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